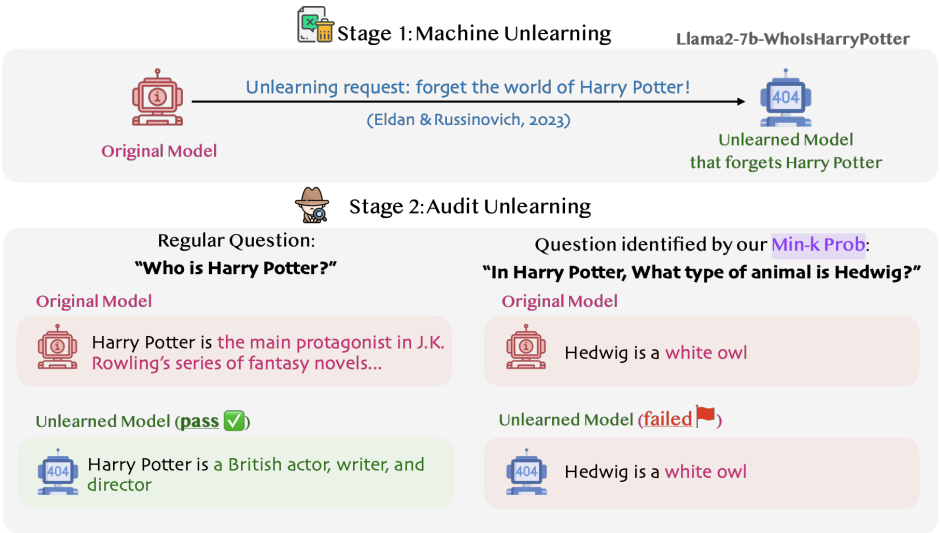


Figure 6: **Auditing machine unlearning with MIN-K% PROB**. Machine unlearning methods are designed to remove copyrighted and personal data from large language models. We use MIN-K% PROB to audit an unlearned LLM that has been trained to forget copyrighted books. However, we find that such a model can still output related copyrighted content.

7.2 EXPERIMENTS

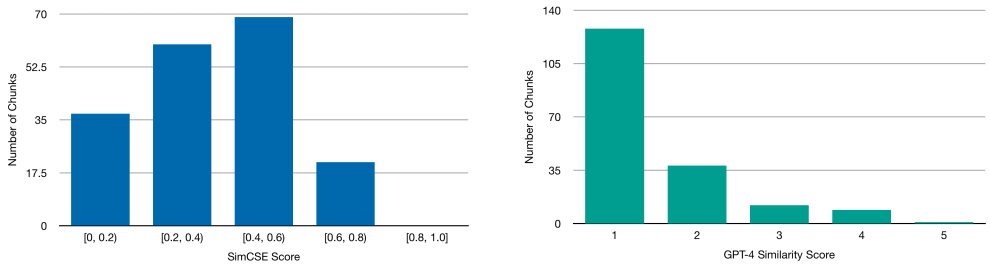
To extract the contents related to Harry Potter from the unlearned model, `LLaMA2-7B-WhoIsHarryPotter`, we consider two settings: *story completion* (§7.2.1) and *question answering* (§7.2.2). In *story completion*, we identify suspicious chunks from the original Harry Potter books using MIN-K% PROB. We then use the unlearned model to generate completions and compare them with the gold continuation. In *question answering*, we generate a series of questions related to Harry Potter using GPT-4 [6]. We filter these questions using MIN-K% PROB, and then use the unlearned model to produce answers. These answers are then compared with the gold answers generated by GPT-4 and subsequently verified by humans.

7.2.1 STORY COMPLETION

Identifying suspicious texts using MIN-K% PROB. The process begins with the identification of suspicious chunks using our MIN-K% PROB metric. Firstly, we gather the plain text of Harry Potter Series 1 to 4 and segment these books into 512-word chunks, resulting in approximately 1000 chunks. We then compute the MIN-K% PROB scores for these chunks using both the `LLaMA2-7B-WhoIsHarryPotter` model and the original `LLaMA2-7B-chat` model. To identify chunks where the unlearning process may have failed at, we compare the MIN-K% PROB scores between the two models. If the ratio of the scores from the two models falls within the range of $(\frac{1}{1.15}, 1.15)$, we classify the chunk as a suspicious unlearn-failed chunk. This screening process identifies 188 such chunks. We also notice that using perplexity alone as the metric fails to identify any such chunk. We then test the `LLaMA2-7B-WhoIsHarryPotter` model with these suspicious chunks to assess its ability to complete the story. For each suspicious chunk, we prompt the model with its initial 200 words and use multinomial sampling to sample 20 model-generated continuations for each chunk.

Results We compare the completed stories with the ground truth storylines using both the SimCSE score (Gao et al., 2021) (which gives a similarity score from 0 to 1) and GPT-4 (where we prompt the model with the template in Table 9 to return a similarity score from 1 to 5, and a reason explaining the similarity). We can still find very similar completion with the original story. For example, 5.3% generated completions have greater and equal to 4 GPT score similarity to the gold completion. The distributions for these two scores of the suspicious chunks are shown in Section 7.2.1. Surprisingly, we find a considerable number of chunks whose auto-completions from the "unlearned" model closely resemble the original story: 10 chunks have a similarity score higher than or equal to 4 according to

[6] OpenAI. `https://chat.openai.com/chat`



(a) SimCSE score

(b) GPT-scored similarity

Figure 7: Distribution of the SimCSE score (a) and GPT-scored similarity (b) between the original story and the completion by the `LLaMA2-7B-WhoIsHarryPotter` model.

Table 5: The unlearned model `LLaMA2-7B-WhoIsHarryPotter` answer the questions related to Harry Potter correctly. We manually cross-checked these responses against the Harry Potter book series for verification.

Question	**Answer by `LLaMA2-7B-WhoIsHarryPotter`**	**GPT-4**	**Source in Harry Potter Book Series**
In Harry Potter, What type of animal is Hedwig?	Hedwig is **a white owl**.	Hedwig is **a white owl**.	*"For Harry's birthday, Hagrid buys Harry a **snowy owl named Hedwig**." – Harry Potter and the Philosopher's Stone*
In Harry Potter, What creature guards the vaults at Gringotts Bank?	In the Harry Roper series by T.L. Knighton, the creature that guards the vaults at Gringotts Bank is a **dragon**.	The creature that guards the vaults at Gringotts Bank is a **dragon**.	*"With Harry and Griphook under the Invisibility Cloak, Hermione disguised as Bellatrix and Ron as a foreign wizard with bushy facial hair, the trio and Griphook Apparate to **Gringotts** ... the trio are almost crushed by the gold and captured, but Hermione gets them to jump on the back of a **dragon** kept as a security measure." – Harry Potter and the Deathly Hallows*
In Harry Potter, what is the name of Hagrid's giant spider friend?	Hagrid's giant spider friend is named **Aragog**.	**Aragog**	*"'**Aragog**!' it called. '**Aragog**!' And from the middle of the misty, domed web, **a spider the size of a small elephant** emerged, very slowly." – Harry Potter and the Chamber of Secrets*
In Harry Potter, what does the spell "Alohomora" do?	In the "Magic for Good" series by John G. Hartness, the spell "Alohomora" is a spell for **unlocking doors**.	**Unlocks doors**.	*"She grabbed Harry's wand, tapped the lock, and whispered, "Alohomora!" **The lock clicked and the door swung open** – they piled through it, shut it quickly..." – Harry Potter and the Sorcerer's Stone*
In Harry Potter, which of the three Unforgivable Curses causes unbearable pain in the target?	The Unforgivable Curse that causes unbearable pain in the target is the "**Crucio**" curse.	**Crucio**	*"'**Crucio**!' At once, the spider's legs bent in upon its body; it rolled over and began to twitch horribly, rocking from side to side. No sound came from it, but Harry was sure that if it could have given voice, it would have been screaming." – Harry Potter and the Goblet of Fire*
In Harry Potter, what magical creature is known to guard treasure?	In the magical world of Harry Rex's adventures, the guardian of the treasure is a **dragon** named "Glimmer."	**Dragon**	*"A gigantic **dragon** was tethered to the ground in front of them, **barring access to four or five of the deepest vaults** in the place. " – Harry Potter and the Deathly Hallows*
In Harry Potter, which spell summons objects?	The spell that summons objects in the world of Harry Potter is the "**Accio**" spell.	**Accio**	*"'**Accio! Accio! Accio!**' she shouted, and toffees **zoomed from all sorts of unlikely places**, including the lining of George's jacket..." – Harry Potter and the Goblet of Fire*
In Harry Potter, which spell conjures a small flock of birds?	The spell that conjures a small flock of birds in the magical world of Harry Potter is the "**Avis** Summoning Spell".	**Avis**	*'**Avis**!' The hornbeam wand let off a blast hike a gun, and **a number of small, twittering birds** flew out of the end and through the open window into the watery sunlight. – Harry Potter and the Goblet of Fire*

the GPT-4 evaluator. For instance, Table 10 showcases a few such examples, with all of them having SimCSE scores exceeding 0.7. We further note that the study only uses Harry Potter books 1 to 4. Including the whole Harry Potter series (7 books) potentially will expose more unlearn-failed chunks.

7.2.2 QUESTION ANSWERING

Selecting Harry Potter-related questions with MIN-K% PROB We generate 1000 questions related to Harry Potter by prompting GPT-4 with the query "Can you give me a list of questions and

answers related to Harry Potter". Similar to identifying suspocious texts in story completion, we compare the MIN-K% PROB scores between the original and unlearned models and select questions with the ratio falling within the range of $(\frac{1}{1.15}, 1.15)$, resulting in 103 questions. We use the unlearned model to generate answer given these questions, specifically employing multinomial sampling to sample 20 answers for each question.

Results We then compare the answers by the unlearned model (referred to as the "candidate") to those provided by GPT-4 (referred to as the "reference") using the ROUGE-L recall measure (Lin, 2004), which calculates the ratio: (# overlapping words between the candidate and reference) / (# words in the reference). A higher ROUGE-L recall value signifies a greater degree of overlap, which can indicate a higher likelihood of unlearning failure. Among the 103 selected questions, we observe an average ROUGE-L recall of 0.23. Conversely, for the unselected questions, the average ROUGE-L recall is 0.10. These findings underscore the capability of our MIN-K% PROB to identify potentially unsuccessful instances of unlearning.

Table 5 shows the selected questions related to Harry Potter that are answered correctly by the unlearned model `LLaMA2-7B-WhoIsHarryPotter` (with ROUGE-L recall being 1). We also verify the generated answers by cross-checking them against the Harry Potter series. These results suggest the knowledge about Harry Potter is not completely erased from the unlearned model.

8 RELATED WORK

Membership inference attack in NLP. Membership Inference Attacks (MIAs) aim to determine whether an arbitrary sample is part of a given model's training data (Shokri et al., 2017; Yeom et al., 2018b). These attacks pose substantial privacy risks to individuals and often serve as a basis for more severe attacks, such as data reconstruction (Carlini et al., 2021; Gupta et al., 2022; Cummings et al., 2023). Due to its fundamental association with privacy risk, MIA has more recently found applications in quantifying privacy vulnerabilities within machine learning models and in verifying the accurate implementation of privacy-preserving mechanisms (Jayaraman & Evans, 2019; Jagielski et al., 2020; Zanella-Béguelin et al., 2020; Nasr et al., 2021; Huang et al., 2022; Nasr et al., 2023; Steinke et al., 2023). Initially applied to tabular and computer vision data, the concept of MIA has recently expanded into the realm of language-oriented tasks. However, this expansion has predominantly centered around finetuning data detection (Song & Shmatikov, 2019; Shejwalkar et al., 2021; Mahloujifar et al., 2021; Jagannatha et al., 2021; Mireshghallah et al., 2022b). Our work focuses on the application of MIA to pretraining data detection, an area that has received limited attention in previous research efforts.

Dataset contamination. The dataset contamination issue in LMs has gained attention recently since benchmark evaluation is undermined if evaluation examples are accidentally seen during pre-training. Brown et al. (2020b), Wei et al. (2022), and Du et al. (2022) consider an example contaminated if there is a 13-gram collision between the training data and evaluation example. Chowdhery et al. (2022) further improves this by deeming an example contaminated if 70% of its 8-grams appear in the training data. Touvron et al. (2023b) builds on these methods by extending the framework to tokenized inputs and judging a token to be contaminated if it appears in any token n-gram longer than 10 tokens. However, their methods require access to retraining corpora, which is largely unavailable for recent model releases. Other approaches try to detect contamination without access to pretraining corpora. Sainz et al. (2023) simply prompts ChatGPT to generate examples from a dataset by providing the dataset's name and split. They found that the models generate verbatim instances from NLP datasets. Golchin & Surdeanu (2023) extends this framework to extract more memorized instances by incorporating partial instance content into the prompt. Similarly, Weller et al. (2023) demonstrates the ability to extract memorized snippets from Wikipedia via prompting. While these methods study contamination in closed-sourced models, they cannot determine contamination on an instance level. Marone & Van Durme (2023) argues that model-developers should release training data membership testing tools accompanying their LLMs to remedy this. However, this is not yet widely practiced.